

Candidate Name: _____

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MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2016
Higher 2

H2 BIOLOGY

9648/01

Paper 1 Multiple Choice Questions

22 September 2016**1 hour 15 minutes**Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencils.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Write your name, civics group and index number on the Multiple Choice Answer Sheet provided.

There are **forty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

You may keep this booklet after the examination.

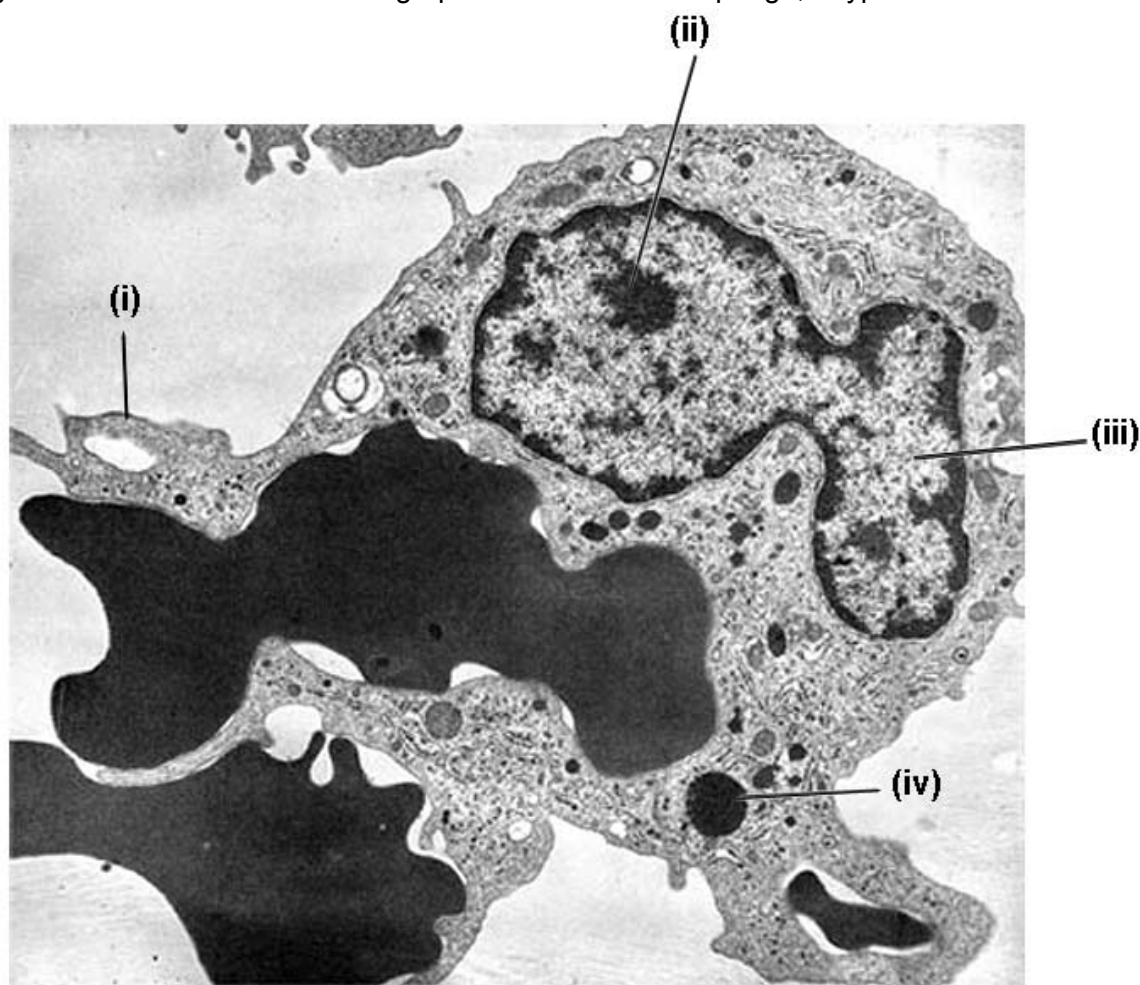
A student used centrifugation to separate the various intracellular structures of human liver cells by size and density. Which of the following molecule(s) would you expect to find in the fraction containing the mitochondria?



- JC2 Prelim Exam 2016 Biology 9648/01

QUESTION 2

The figure below is an electron micrograph of a human macrophage, a type of white blood cell.



Which of the options correctly matches structure with function?

	Structure (i)	Structure (ii)	Structure (iii)	Structure (iv)
A.	Engulfs foreign bacteria	Contains genes that code for hydrolytic enzymes	Transcription of ribosomal RNA	Contains enzymes for secretion
B.	Engulfs worn out red blood cells	Partial assembly of ribosomes	Contains genes that code for specific receptor proteins	Contains hydrolytic enzymes
C.	Engulfs foreign bacteria	Transcription of ribosomal RNA	Contains genes that code for specific receptor proteins	Contains enzymes for secretion
D.	Engulfs worn out red blood cells	Contains genes that code for hydrolytic enzymes	Transcription of ribosomal RNA	Contains hydrolytic enzymes

QUESTION 3

The phospholipid bilayer of a certain type of cell was analysed by separating the two layers and analysing the components of each layer.

Which option shows the composition of each layer?

A.	Glycolipids	Phospholipids	Glycoproteins	Cholesterol
Inner layer	0%	80%	0%	20%
Outer layer	15%	50%	15%	20%

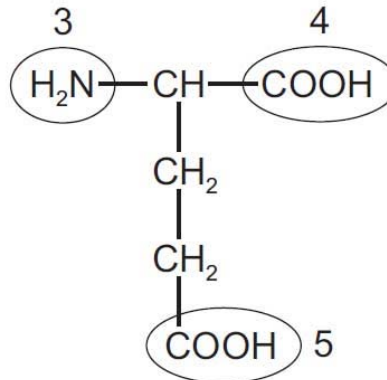
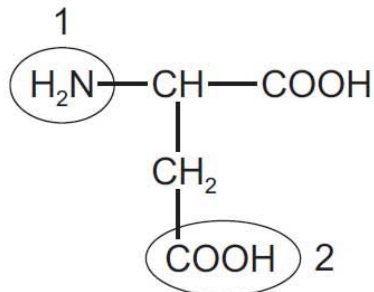
B.	Glycolipids	Phospholipids	Glycoproteins	Cholesterol
Inner layer	15%	50%	15%	20%
Outer layer	0%	80%	0%	20%

C.	Glycolipids	Phospholipids	Glycoproteins	Cholesterol
Inner layer	10%	60%	20%	10%
Outer layer	30%	50%	0%	20%

D.	Glycolipids	Phospholipids	Glycoproteins	Cholesterol
Inner layer	30%	50%	0%	20%
Outer layer	15%	50%	15%	20%

QUESTION 4

The diagrams show the structures of two amino acids, each of which has two carboxylic acid groups ($-\text{COOH}$).



Which groups form the bonds that maintain the configuration of α -helices?

- A. 1 and 4 B. 1 and 5 C. 2 and 3 D. 2 and 5

QUESTION 5

Which features adapt a cellulose molecule for its function?

1. Long chains of β -glucose molecules have multiple branches.
2. Many hydrogen bonds are formed between adjacent chains.
3. Cellulose is insoluble in water.

- A. 1, 2 and 3 B. 1 and 3 only C. 2 and 3 only D. 2 only

QUESTION 6

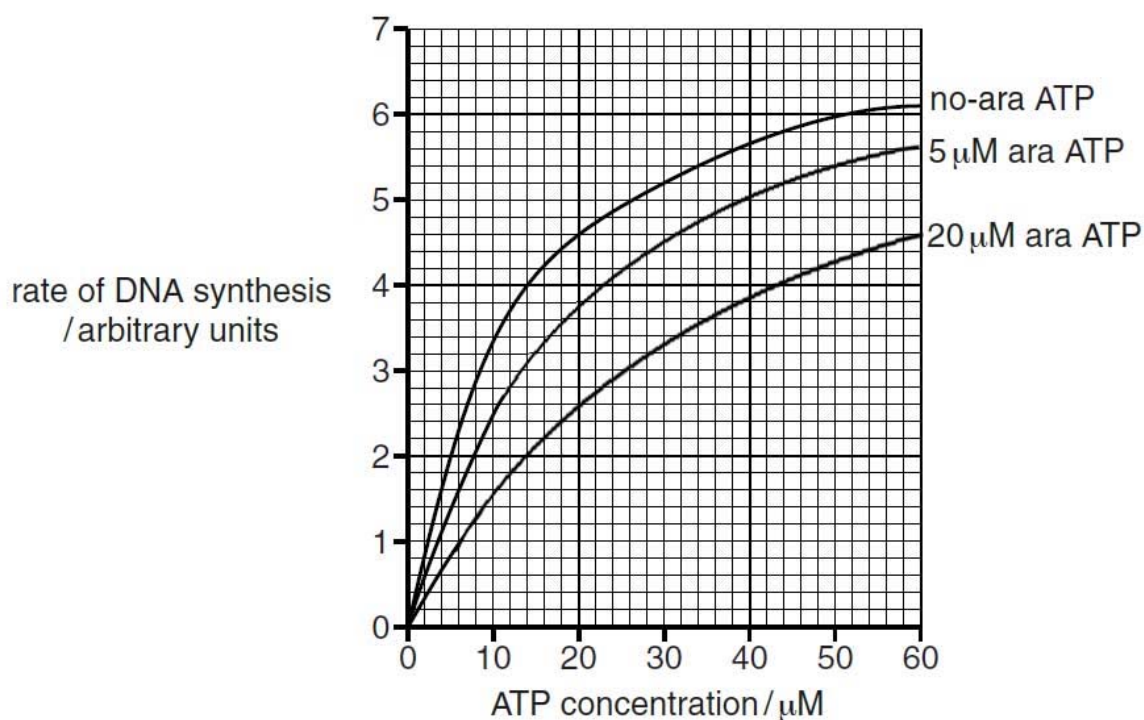
DNA polymerase is an enzyme involved in the replication of DNA. One of the substrates required by DNA polymerase is ATP.

ara-ATP is a chemical that affects DNA polymerase activity.

In an investigation, the effect of different concentrations of ATP on the rate of DNA synthesis was determined:

- with no ara-ATP
- with a low concentration of ara-ATP
- with a high concentration of ara-ATP

The results of the investigation are shown in the graph below:



Which of the following statements about the effects of ara-ATP are **false**?

1. ara-ATP binds to an allosteric site on DNA polymerase.
2. ara-ATP binds to the active site on DNA polymerase.
3. ara-ATP is similar in structure to ATP.
4. When ara-ATP binds to DNA polymerase, the shape of its active site changes.
5. When ara-ATP binds to DNA polymerase, the rate of DNA synthesis can be increased by increasing the concentration of ATP.

A. 1 and 4

B. 1 and 5

C. 2 and 3

D. 2 and 5

QUESTION 7

Which is the correct statement concerning cell and nuclear division?

- A. At prophase, the mass of DNA is doubled. Following anaphase, this mass is reduced by half and following cytokinesis this mass halves again.
- B. Mutagens can cause mutations whereas carcinogens can cause cancer. This means that all mutagens are carcinogenic.
- C. Some of the roles of mitosis are growth, asexual reproduction, cell repair following tissue damage and cell replacement.
- D. Haploid eukaryotes can reproduce by mitosis whereas diploid eukaryotes can reproduce by mitosis or meiosis.

QUESTION 8

Some plants, such as wheat or banana plants, can produce diploid or haploid gametes. These gametes can fertilise other diploid or haploid gametes.

Which statements are correct for plants like these?

1. Diploid gametes may be produced by non-disjunction during meiosis.
2. The offspring will always show an increased chromosome number.
3. The offspring could be $2n$, $3n$ or $4n$.
4. The chromosome number could increase with each generation.

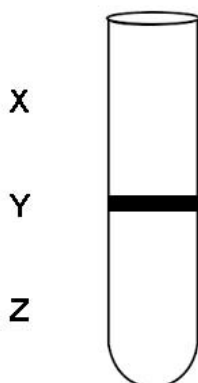
- A. 1, 2 and 3 B. 1, 2 and 4 C. 1, 3 and 4 D. 2, 3 and 4

QUESTION 9

A culture of bacteria was allowed to reproduce using nucleotides containing ^{14}N for many generations. The culture was then allowed to reproduce using nucleotides with the heavy isotope of nitrogen, ^{15}N , for one generation. The DNA of the bacterial cells was then examined using a centrifuge before it was returned to a culture medium with nucleotides containing ^{14}N .

The DNA of the bacterial cells was then examined again after two subsequent generations in the culture medium with nucleotides containing ^{14}N .

The diagram below shows the position of the DNA band at Y in the centrifuge tube when the DNA was first labelled.

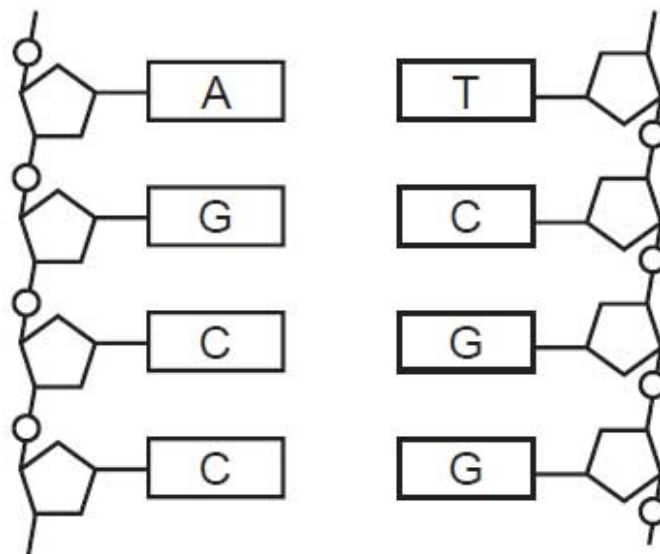


Which option shows the number of bands and their respective band positions for the two subsequent generations in the culture medium with nucleotides containing ^{14}N .

	After one generation in ^{14}N medium	After another generation in ^{14}N medium
A.	Two bands, 50% at Y and 50% at Z	Two bands, 75% at Y and 25% at Z
B.	Two bands, 50% at Y and 50% at Z	Two bands, 25% at Y and 75% at Z
C.	Two bands, 50% at X and 50% at Y	Two bands, 75% at X and 25% at Y
D.	Two bands, 50% at X and 50% at Y	Two bands, 25% at X and 75% at Y

QUESTION 10

The diagram shows part of a DNA molecule.



How many hydrogen bonds are involved in holding these strands of DNA together?

- A. 12 B. 11 C. 9 D. 8

QUESTION 11

In 1985, it was discovered that a bacterium, *Mycoplasma capricolum*, used a deviant genetic code. The codon UGA resulted in the addition of tryptophan to the growing polypeptide chain.

A short sequence of nucleotides was synthesised with the following base sequence:

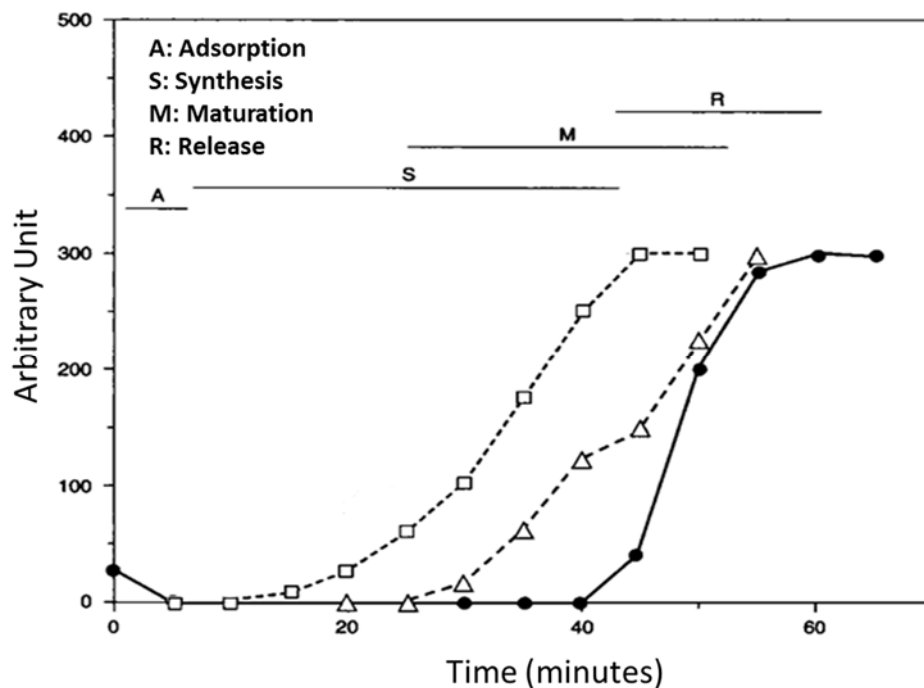
3' CTGGCAACTATTTCAACTCATATC 5'

How many peptide bonds would be formed by ribosomes when this sequence under goes transcription and translation in *Mycoplasma capricolum* and a human liver cell?

	<i>Mycoplasma capricolum</i>	Human liver cell
A.	1	0
B.	2	1
C.	3	2
D.	4	3

QUESTION 12

The graph represents the process of viral replication.



Which of the following option is correct?

	●	□	△
A.	extracellular virions	nucleic acids and proteins	intracellular viral particles
B.	extracellular virions	nucleic acids	intracellular viral particles
C.	intracellular viral particles	nucleic acids and proteins	extracellular virions
D.	intracellular viral particles	nucleic acids	extracellular virions

QUESTION 13

Which of the following correctly describe drugs that are effective in treating viral infection?

1. Antibiotics that induce the body to produce antibodies
2. Drugs that interfere with the synthesis of viral nucleic acid
3. Drugs that prevent viral protein synthesis by interfering with viral ribosomes
4. Drugs that change the cell surface receptor on the host cell
5. Drugs that prevent uncoating of the nucleocapsids

A. 2 and 5

B. 4 and 5

C. 2 and 3

D. 1 and 3

QUESTION 14

Which of the following statements is true for generalized and specialized transduction?

	Generalized	Specialized
A.	Transfers any bacterial DNA	Transfers one specific bacterial gene
B.	Contains a hybrid DNA in its capsid	Contains only bacterial DNA in its capsid
C.	The host cell will die	The host cell will not die
D.	Viral DNA is replicated by host cell machinery	Viral DNA is replicated by binary fission

QUESTION 15

Which statements about bacterial genetic transfer are **not** correct?

1. In transformation, bacterial cells which possess competence factors can only take up the plasmids from the surroundings.
2. Homologous recombination is always involved in bacterial genetic transfer.
3. After conjugation, the donor and recipient cells contain the same genetic information.
4. Binary fission will not contribute to genetic variation in bacterial cells without plasmids.

A. 1 and 3 **B.** 1, 2 and 3 **C.** 2 and 3 **D.** 1, 2, and 4

QUESTION 16

Which statement describes the difference between an inducible and repressible operon?

	Inducible operon	Repressible operon
A.	Functions in catabolic pathways	Functions in anabolic pathways
B.	Repressor genes are usually not expressed	Repressor genes are usually expressed
C.	Synthesizes inactive repressor	Synthesizes active repressor
D.	Repressor protein activated by substrate	Repressor protein repressed by substrate

QUESTION 17

In 1979, six groups of investigators independently reported the discovery of a p53 protein (encoded by *TP53* gene) that was present in human and mouse cells. One of the groups discovered that the p53 protein level was highly expressed in several types of mouse tumour cells. In 1980s, another research group discovered the association of high p53 protein level with human intestinal tumour cells but not normal cells.

What conclusion did the scientists arrive at based on the information above?

- A. *TP53* is a tumour suppressor gene
- B. *TP53* is a proto-oncogene
- C. p53 is a transcription factor
- D. *TP53* is not expressed in normal cells

QUESTION 18

Which statement concerning polypeptide synthesis is correct?

- A. A particular cell type will transcribe all the genes present in one set of chromosomes but will only process particular pre-mRNA transcripts to enable polypeptide synthesis.
- B. Different cell types contain different sets of genes to produce different pre-mRNA transcripts and synthesise different polypeptides.
- C. The same pre-mRNA transcripts are synthesised by all cell types but different introns are removed from the transcripts before translation to synthesise polypeptides.
- D. Different polypeptides can be synthesized from the same pre-mRNA in different cell types.

QUESTION 19

Gene expression in eukaryotes can be regulated at the translational level.

Which combination of statements correctly describes eukaryotic translational control?

	Condition	Effect
A.	Lack of translation initiation factor proteins	Inhibition of translation of selected mRNA
B.	Presence of repressor proteins binding to 5'-UTR of selected mRNA	Prevents small ribosomal subunit from binding
C.	Presence of repressor proteins to distal control elements	mRNA is translationally-repressed
D.	mRNA with a long poly-A tail	mRNA is degraded slowly by restriction endonuclease

QUESTION 20

In fruit flies, one gene controls wing form (normal or vestigial) and one gene controls eye colour (red or normal brown).

A fly with normal wings and normal brown eyes is crossed with a fly with vestigial wings and red eyes. All the F_1 are normal for both characteristics.

However, when F_1 are crossed with each other, the resulting F_2 is:

45 normal wing, normal brown eye
17 normal wing, red eye
16 vestigial wing, normal brown eye
5 vestigial wing, red eye
1 normal wing, orange eye

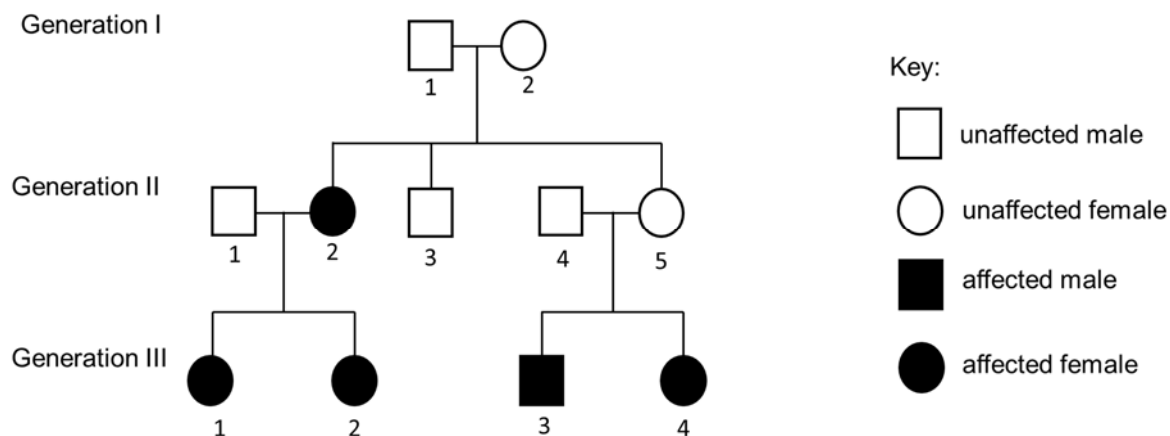
What is the best explanation for the results of this dihybrid cross?

- A. Codominance
- B. Gene mutation
- C. Multiple alleles
- D. Sex linkage

QUESTION 21

Multiple Sclerosis (MS) is a neurodegenerative disease in which the immune system attacks the myelin that protects nerve fiber, upsetting the flow of information between the brain and the body.

The pedigree chart below shows the pattern of MS inheritance in a family.



Which of the following states the inheritance pattern of MS?

- A. Sex-linked dominant
- B. Sex-linked recessive
- C. Autosomal dominant
- D. Autosomal recessive

QUESTION 22

Two parents have a son who has blood group **O** and haemophilia. One parent has blood group **O** and the other has blood group **B**. Neither parent has haemophilia.

What is the probability that the second child of these parents is a son with blood group **B** who does not have haemophilia?

- A. 1 in 4 B. 1 in 8 C. 2 in 4 D. 3 in 8

QUESTION 23

Feathers in poultry can be white or coloured and this is controlled by two genes, **P/p** and **Q/q**. The phenotypes of offspring that are expected from mating two birds, each of which is heterozygous at both loci, are shown in the Punnett square.

gametes	PQ	Pq	pQ	pq
PQ	white feathers	white feathers	white feathers	white feathers
Pq	white feathers	white feathers	white feathers	white feathers
pQ	white feathers	white feathers	coloured feathers	coloured feathers
pq	white feathers	white feathers	coloured feathers	white feathers

Which of the following best explains the proportion of white to coloured feathers in the Punnett square?

- A. Dominant epistasis in which a suppressor prevents the expression of epistatic gene.
B. Dominant epistasis in which the epistatic allele is **P**.
C. Recessive epistasis in which colour is recessive to no colour at one allelic pair.
D. Recessive epistasis in which the epistatic allele is **p**.

QUESTION 24

Coat colour in rabbits is controlled by a gene with four alleles. The order of dominance for these alleles is as follows:

agouti (**C**) > chinchilla (**C^c**) > himalayan (**C^h**) > albino (**c**)

What is the maximum number of different coat colours obtained from a cross between an agouti rabbit and a Himalayan rabbit?

- A. 1 B. 2 C. 3 D. 4

QUESTION 25

Which of the following statements is true?

- A. Continuous variation shows a normal distribution and is only influenced by genetic factors.
- B. Continuous variation is controlled by many genes and the traits are usually well defined with no gradation.
- C. Discontinuous variation shows traits that follow discrete distribution and is mostly influenced by environmental factors.
- D. Discontinuous variation shows traits that are controlled by one or two genes and is relatively unaffected by environmental factors.

QUESTION 26

Scorpions have a pair of grasping claws at the front of their bodies and a tail with a stinger. The stinger is used to inject venom into their prey to cause paralysis and convulsion.

Scorpion venom contains two active components:

- a toxin that affects ion channels at synapse of the nervous system of their prey
- an inhibitor of an enzyme found at these synapses

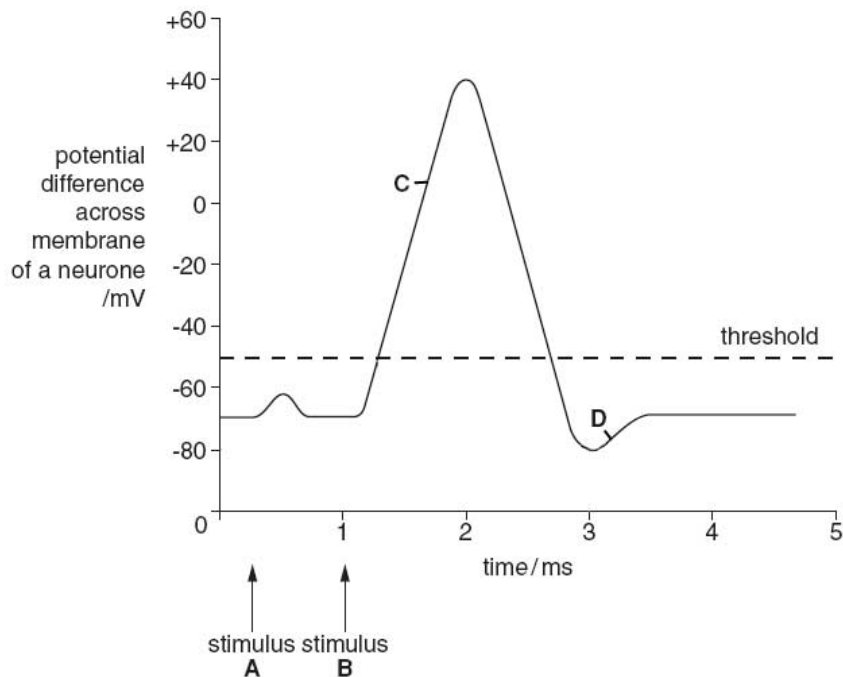
Which of the following statements **incorrectly** explain how the scorpion venom may stop the functioning of the synapse?

1. The toxin prevents the opening of the voltage-gated Na^+ channel at the presynaptic membrane while the inhibitor will cause the opening of the ligand-gated Na^+ channel.
2. Toxin will stop the release of the neurotransmitter into synaptic cleft and inhibitor will stop the recycling of neurotransmitter.
3. Toxin will prevent the entry of Na^+ ions into the postsynaptic knob and inhibitor will allow continuous depolarisation of postsynaptic membrane.
4. Both the toxin and inhibitor do not result in depolarisation of postsynaptic membrane.

- A. 1 and 2 B. 2 and 3 C. 1 and 4 D. 3 and 4

QUESTION 27

The diagram shows an action potential.



1. Stimulus **B** is a stronger stimulus than stimulus **A** and will open more voltage-gated Na^+ channels to overcome the threshold potential.
2. At point **C**, the axon membrane is most permeable to sodium ions.
3. At point **D**, sodium conductance changes more slowly than potassium conductance.
4. A strong intensity stimulus can initiate a second action potential at point **D**.

How many of the above statement(s) is/are correct?

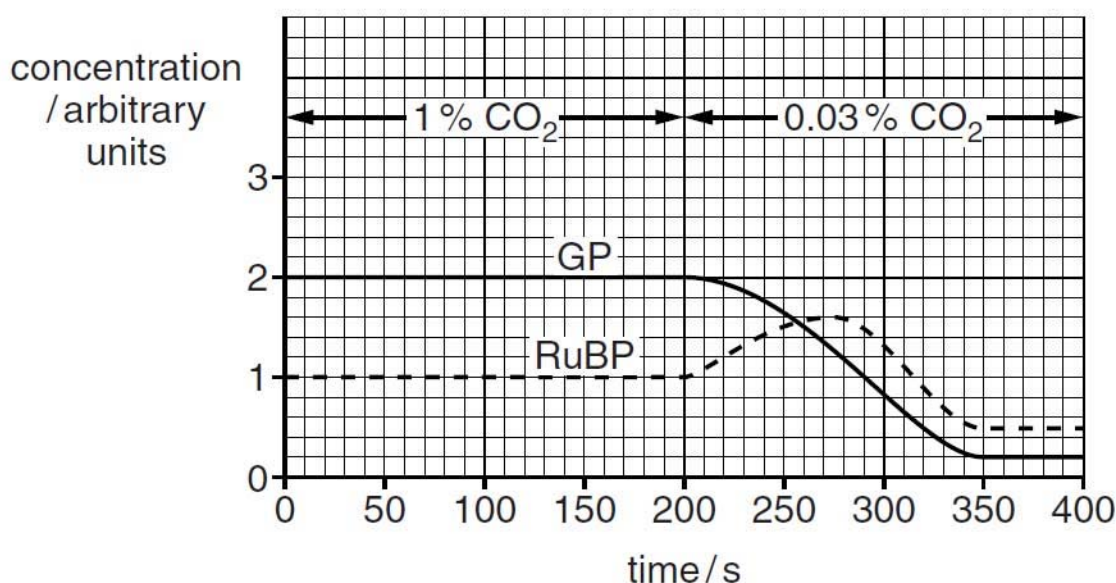
- A.** 1 **B.** 2 **C.** 3 **D.** 4

QUESTION 28

In an experiment, carbon dioxide concentration was altered to investigate its effects on the light-independent stage of photosynthesis.

- A cell suspension of *Chlorella* was illuminated using a bench lamp.
- The suspension was supplied with carbon dioxide at a concentration at 1% for 200 seconds.
- The concentration of carbon dioxide was then reduced to 0.03% for a further 200 seconds.
- The concentrations of RuBP (ribulose biphosphate) and GP (glycerate-3-phosphate) were measured at regular intervals.
- Throughout the investigation the temperature of the suspension was maintained at 25°C.

The results are shown below.



Which of the following statements is/are correct?

1. At 0.03% of CO_2 , concentration of GP decreases due to the decrease in the rate of carbon fixation.
2. The concentration of RuBP increases between 210s and 250s due to more CO_2 fixation and more RuBP regenerated from triose phosphate.
3. There is an accumulation of triose phosphate between 250s to 290s.
4. There is an accumulation of RuBP in the chloroplast between 210s and 250s.

- A. 1 and 4 B. 1 only C. 2 only D. 3 and 4

QUESTION 29

Both glucose and appropriate enzymes are necessary for the process of glycolysis to begin.

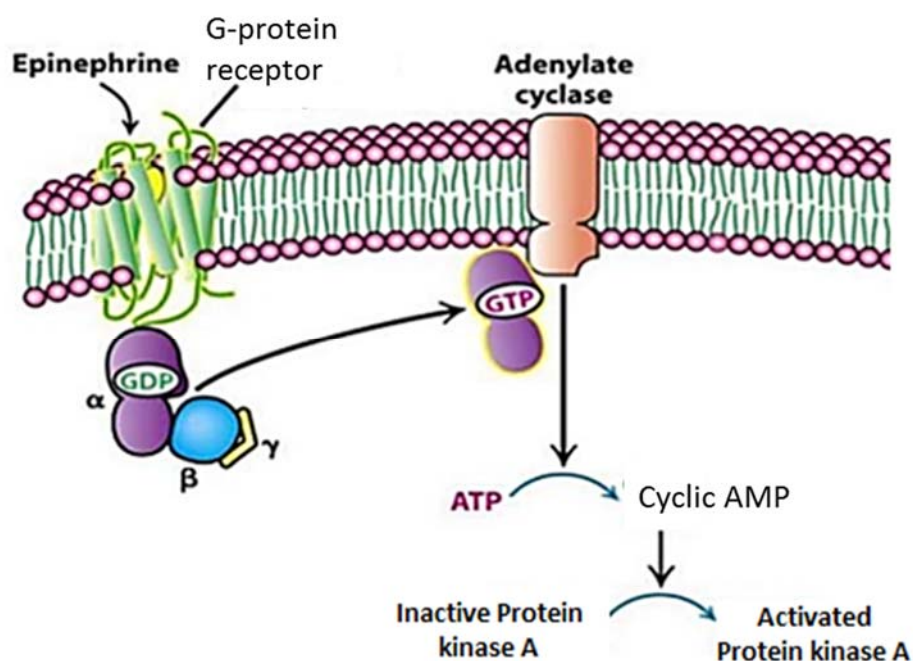
Which additional compound must also be present?

- A. acetyl coenzyme A
- B. ATP
- C. pyruvate
- D. reduced NAD

QUESTION 30

The G protein-coupled receptor is activated by the binding of epinephrine to the receptor.

A mutation leads to constitutive signal transduction.



Which of the following explanations of the mutation are **incorrect**?

1. adenylate cyclase cannot bind to activated GTP
2. conformation change in G-protein receptor causes epinephrine to bind tightly
3. GTPase in G protein cannot hydrolyse GTP to GDP
4. conformation change in adenylate cyclase prevents the conversion of ATP to cyclic AMP

- A. 1 and 3
- B. 3 and 4
- C. 1 and 4
- D. 2 and 3

Question 31

Lucilia cuprina, the sheep blowfly, lays its eggs in wounds and the wet fleece of sheep. The larvae hatch and burrow into the sheep's skin, reduced wool production and sometimes cause death. Particular chemicals were used in the past to control the *L. cuprina* but these became less effective as sheep blowfly developed a resistance to the chemicals.

The cause of the increased resistance to the chemicals was most likely due to

- A. farmers successively reducing the levels of insecticide applied to sheep.
- B. the insecticide producing a change in a gene which enhanced the survival of the blowfly.
- C. a chance mutation in a blowfly gene conferring a survival advantage in the chemical environment.
- D. the insecticide producing a change in phenotype which enhanced reproduction of the blowfly.

Question 32

Which of the following options is correct?

	Factors that are important for a species to evolve by natural selection	Evidence for evolution
A.	Environmental change and inbreeding	Homologous structures and selective breeding of domesticated animals
B.	Environmental change and variation	Homologous structures and overproduction of offspring
C.	Variation and inbreeding	Homologous structures and overproduction of offspring
D.	Environmental change and variation	Homologous structures and selective breeding of domesticated animals

QUESTION 33

The following statements relate to molecular phylogenetics.

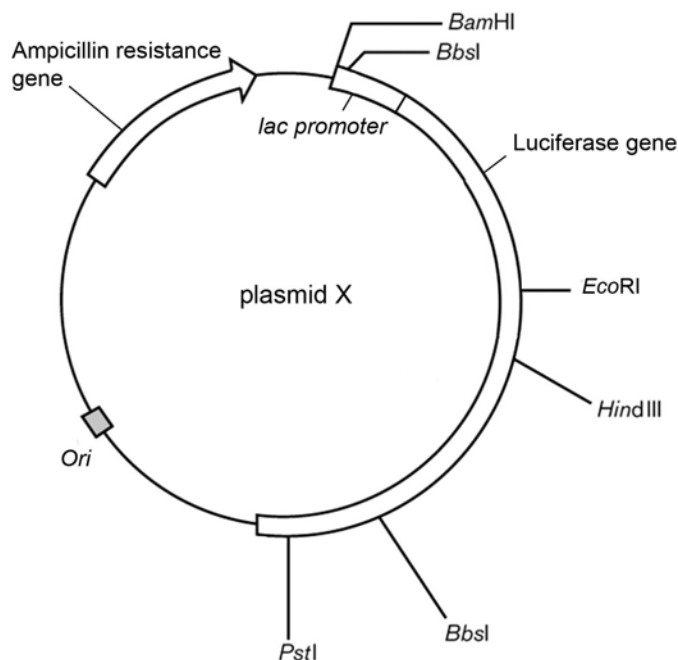
1. Lines of descent from a common ancestor to present-day organisms have undergone fixed rates of accumulation of DNA mutation in any given gene.
2. Organisms with similar base sequences in their DNA are closely related to each other.
3. The number of differences in the base sequences of DNA of different organisms can be used to construct evolutionary trees.
4. Fossil records provide evidence for established periods of evolutionary divergence.

Which statements, when taken together, suggest the existence of a 'molecular clock' that enables scientists to estimate the time at which one species might have diverged from another?

- A. 1 and 2 only B. 1 and 4 only C. 2 and 3 only D. 3 and 4 only

QUESTION 34

Plasmid X can serve as a vector for the insertion of genes to be cloned. Luciferase will allow the bacteria to emit light in presence of luciferin as a substrate.



How many possible restriction sites can be used for the expression of human growth hormone gene?

- A. 1 B. 2 C. 3 D. 5

QUESTION 35

Synthesis of human insulin by genetically-modified bacteria involves the use of the enzyme reverse transcriptase.

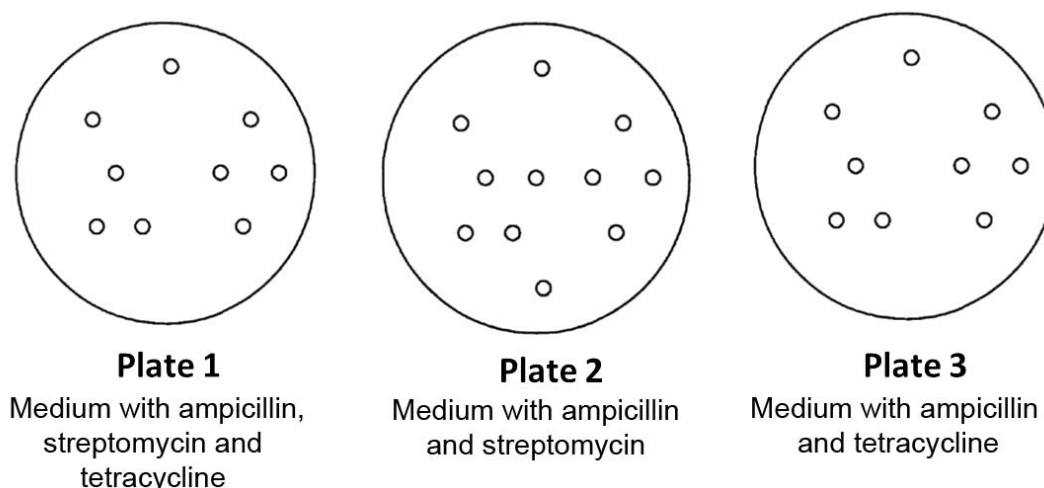
Which of the following statement(s) correctly describe(s) reverse transcriptase used in the above process?

1. It causes complementary DNA to be formed from mRNA
2. It causes single-stranded DNA to be converted to double-stranded DNA
3. It is found in prokaryotic cells
4. It is found in eukaryotic cells

A. 1 only B. 1 and 3 C. 2 and 4 D. 1 and 4

QUESTION 36

A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coli* and the bacteria were grown on a nutrient medium with various antibiotics using replica plating. The resulting plates are shown in the diagram.



Which antibiotic gene(s) contain(s) the restriction site that was used for the insertion of the human gene?

- A. Ampicillin
- B. Streptomycin
- C. Tetracycline
- D. Ampicillin and tetracycline

QUESTION 37

Which of the following statements about PCR are **false**?

1. The PCR cycle involves denaturation of the template, annealing of the RNA primers and polymerization of nucleotides.
2. PCR uses thermostable DNA-dependent DNA polymerases.
3. Magnesium ion ensures the stability of the thermostable DNA polymerases in PCR as it functions as a cofactor for the thermostable DNA polymerases in PCR.
4. Shorter duration of denaturation temperature at 95°C is required if the DNA template has high guanine and cytosine content.

A. 1 and 3 B. 1 and 4 C. 2 and 3 D. 3 and 4

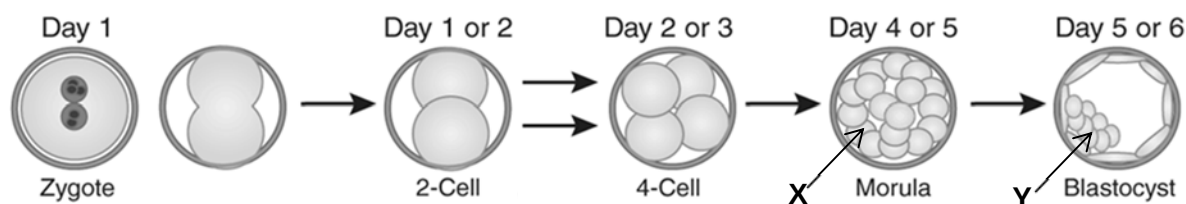
QUESTION 38

Which of the following is true of a genomic library?

- A. It is a collection of the different RFLPs of organisms within a population.
- B. It is a collection of DNA fragments that make up the entire genome of a particular organism.
- C. It is a collection of DNA fragments created by reverse transcriptase which are then inserted into vectors.
- D. It is a collection of all genes of an organism's genome which have been sequenced.

QUESTION 39

The figure below shows several stages in the development of an embryo.



Which of the following statements is true about the cells labelled **X** and **Y**?

- A. **X** is a pluripotent cell while **Y** is a multipotent cell.
- B. **X** is a pluripotent cell while **Y** can give rise to multipotent cells.
- C. **Y** will develop into the entire foetus including its placenta.
- D. **X** can only give rise to totipotent cells but **Y** will give rise to pluripotent cells.

QUESTION 40

What are the possible arguments against the use of genetically modified organisms (GMOs)?

1. Insufficient testing of genetically modified crop for their side effects
2. Unforeseen long-term effects of genetic manipulation
3. Accidental genetic recombination in bacteria present in the lower intestines of humans as a result of consuming food derived from GMOs
4. Control of food supply by a small number of companies that have access to genetic engineering technology

A. 1 and 2

B. 2 and 3

C. 1, 2 and 3

D. 1, 2, 3 and 4

END OF PAPER 1